

Matrix Theory: Lecture 12

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Outline

Hermitian forms

- Congruent Matrices
- Signature
- Completing squares
- Positive definite forms



Hermitian Forms

Def. Let V be a complex vector space. A **Hermitian form** on V is a function of form

$$f(X) = \sum_{i,j} a_{ij} \bar{x}_i x_j, = X^H A X$$

where X is a vector with the (complex!) coordinates x_i w.r.t. to some basis and $A = (a_{ij})$ is a **Hermitian** matrix.

Note that $f(X)$ is a real number since

$$X^H A X = (X, A X) = (A^H X, X) = (A X, X) = \overline{(X, A X)} = \overline{X^H A X}$$

If X, A are real and A is **symmetric** then

$$f(X) = \sum_{i,j} a_{ij} x_i x_j = X^T A X,$$

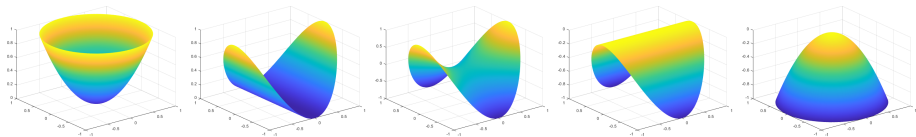
is called a **Quadratic form**.



Hermitian Forms

The Hermitian (in real case the quadratic) form and the corresponding matrix A are called

- **positive definite** if $f(X) > 0$ for $X \neq O$;
- **positive semidefinite** if $f(X) \geq 0$ for all X ;
- **indefinite** if $f(X)$ attains both negative and positive values.
- **negative semidefinite** if $f(X) \leq 0$ for all X ;
- **negative definite** if $f(X) < 0$ for $X \neq O$;



Congruent Matrices

Def. Two Hermitian matrices A and B are **congruent** if $B = S^H A S$ for some invertible matrix S .

If $X = S\hat{X}$ then

Linear operator:	Hermitian form:
$Y = AX$	$f(X) = X^H A X$
$\hat{A} = S^{-1} A S$	$\hat{A} = S^H A S$
A and $\hat{A}$ are similar	A and $\hat{A}$ are congruent



Diagonalization of Hermitian Forms

Theorem 14.3

For every Hermitian form $f(X) = X^H A X$ one can choose a basis such that in this basis

$$f(\hat{X}) = \lambda_1 |\hat{x}_1|^2 + \cdots + \lambda_n |\hat{x}_n|^2.$$

Here λ_i are eigenvalues of the original matrix A . In particular,

- $f(X)$ is positive definite iff all eigenvalues are positive;
- $f(X)$ is positive semidefinite iff all eigenvalues are non-negative;
- $f(X)$ is negative definite iff all eigenvalues are negative;
- $f(X)$ is negative semidefinite iff all eigenvalues are non-positive;
- $f(X)$ is indefinite if it has both positive and negative eigenvalues.



Congruent Matrices

Theorem 14.6

Any Hermitian matrix is congruent to one and only one diagonal matrix of form $\text{diag}(1, 1, \dots, 1, -1, -1, \dots, -1, 0, 0, \dots, 0)$.

If p is the number of ones, m is the number of minus ones (both numbers can be zero) then

$$r(A) = p + m.$$

The numbers p and m are equal to the number of positive and negative d_i in any diagonalization $f(\hat{X}) = \sum d_i |\hat{x}_i|^2$.

In particular p is equal to the number of the positive eigenvalues, m is equal to the number of negative eigenvalues.

The pair (p, m) is called the **signature** of the matrix. Thus two Hermitian matrices of the same size are congruent if and only if they have the same signature.



Diagonalization of Hermitian Forms

$$\Delta_1 = a_{11}, \Delta_2 = \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix}, \Delta_3 = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}, \dots, \Delta_n = \det(A).$$

Theorem 14.4

Let A be Hermitian with all $\Delta_k \neq 0$. Then there exists a lower triangular matrix L with 1 on the main diagonal and a diagonal matrix $D = \text{diag}(d_1, \dots, d_n)$ such that

$$A = LDL^H.$$

If $f(X) = X^HAX$ and $X = (L^{-1})^H \hat{X}$ then $f(\hat{X}) = \sum d_i |\hat{x}_i|^2$.

The matrices L^H and D can be found by the strict Gaussian elimination process.

Moreover, the same can be true if some $\Delta_k = 0$, when the matrix is not indefinite. The process in this case is slightly modified as described above.

In the cases where the modified process cannot be performed there is no such decomposition and the matrix is indefinite.

Positive Definite Hermitian Forms

Theorem 14.6

The following conditions are equivalent for a Hermitian matrix A :

- 1 The matrix is positive definite, thus $f(X) = X^H A X > 0$ for any $X \neq 0$.
- 2 All eigenvalues are positive.
- 3 A is congruent to a diagonal matrix with positive diagonal.
- 4 $A = LDL^H$, where D is a diagonal matrix with positive numbers on the diagonal and L is lower triangular matrix with 1 on the main diagonal.
- 5 All $d_i > 0$ in the process of Gaussian elimination above.
- 6 All minors $\Delta_k > 0$.
- 7 A is congruent to the identity matrix I .
- 8 The signature of the matrix is equal to $(n, 0)$, where n is its size.
- 9 $f(X) = (X, X)$ for some inner product (X, Y) .
- 10 $A = B^H B$ for some invertible matrix B .
- 11 $A = GG^H$, where G is a lower triangular matrix with positive diagonal. (**Cholesky decomposition**)
- 12 $A = P^2$ for some positive definite matrix P . More exactly $P = \sqrt{A}$ – as a function on matrices.